WORKING PAPER — SYSTEMS ANALYSIS

Submitted for Comment | Not Peer Reviewed

You Are Wasting 98% of Your Inference Compute:

54x Intelligence Per Watt from Software Alone — A Multiplicative Algorithmic Stack for LLM Inference

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April 2026

Abstract

Saad-Falcon et al. (2025) introduced Intelligence Per Watt (IPW) as the critical metric for tracking AI efficiency: task accuracy divided by power consumed. Their longitudinal study documents 5.3x IPW improvement from 2023-2025, driven by model and hardware advances. This paper demonstrates that a stack of algorithmic efficiency optimizations — derived from a unified stochastic health monitoring framework — provides an additional multiplicative IPW improvement on top of whatever hardware is available.

Under a partial independence assumption, we estimate an upper bound of 54x IPW improvement at sequence length 4,096 tokens from a three-layer algorithmic stack: per-operation memory bandwidth efficiency (FlashAttention, Dao et al. 2023, 2.86x), adaptive sampling allocation (power metric inference, Paper 2, 13.7x), and depth efficiency (early exit, Paper 8, 1.39x). Each layer targets a structurally distinct compute bottleneck, supporting the independence assumption. These gains are unlikely to fully co-occur in all workloads — the stack represents an upper bound when all three bottlenecks are simultaneously active. Conservative and moderate scenarios are presented in Section 4. Critically, these algorithmic gains are available today on existing hardware — they do not require waiting for the next hardware generation.

Keywords: intelligence per watt, IPW, energy efficiency, algorithmic efficiency, FlashAttention, power metric, early exit, speculative decoding, compute stack, Saad-Falcon

The headline number is large. The appropriate reaction is not acceptance or dismissal — it is calibration. The 54x figure is a structured upper bound, not a deployment guarantee. Real-world gains depend on workload characteristics, layer overlap, and implementation quality. The purpose of this paper is to map the hypothesis space and identify where empirical validation is most valuable. Even 10% of 54x is 5.4x — material on any infrastructure running millions of queries daily.

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1. Introduction

The exponential growth in AI inference demand is creating unprecedented pressure on computational infrastructure. Saad-Falcon et al. (2025) document this pressure and propose Intelligence Per Watt as the unifying metric: how much accurate intelligence does a system produce per unit of energy consumed? Their study finds 5.3x IPW improvement from 2023-2025, driven equally by better models and better hardware.

Their paper focuses on measuring IPW across hardware platforms and tracking its evolution. It does not address the algorithmic efficiency layer — the question of how software-level optimizations can improve IPW independent of hardware. This paper addresses that question.

We identify three algorithmic layers that each independently improve IPW and together compound multiplicatively under a partial independence assumption. The independence assumption is the key constraint: FlashAttention targets memory bandwidth, power metric inference targets sampling compute, and early exit targets depth compute — these are structurally distinct axes that do not obviously compete for the same bottleneck. However, these gains are unlikely to fully co-occur in all workloads. The stack represents an upper bound when all three bottlenecks are simultaneously active.

The relationship to Saad-Falcon et al.'s work is additive, not competitive. Their 5.3x came from better chips and better model architectures. Our 54x core stack comes from better allocation of whatever compute those chips provide. On 2025 hardware, both apply simultaneously. The combined improvement is their product.

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2. IPW and the Algorithmic Layer

2.1 Intelligence Per Watt Defined

Saad-Falcon et al. define IPW as:

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IPW = task_accuracy / power_consumed
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where power is measured in watts and accuracy is measured on a standardized benchmark suite across real-world queries. The metric captures both the capability of the system (numerator) and its energy cost (denominator). A system with higher IPW delivers more correct answers per unit of energy.

IPW can be improved in three ways: improve accuracy, reduce power, or both. Hardware improvements primarily reduce power (more efficient chips, better memory bandwidth). Model improvements primarily improve accuracy (better architectures, better training). Algorithmic efficiency improvements target both: some reduce the power consumed per inference step, others prevent accuracy degradation that would lower the numerator.

2.2 Why Algorithmic Gains Are Orthogonal to Hardware Gains

Saad-Falcon et al.'s hardware gains are fixed by the physical properties of the chip. On any given hardware platform, IPW is maximized by (a) choosing the most capable model and (b) running it as efficiently as possible. Their paper measures (a). This paper addresses (b).

Algorithmic efficiency operates at a different layer than hardware efficiency. FlashAttention reduces memory reads/writes per attention operation — a software optimization that applies

regardless of GPU generation. The power metric reduces which training and inference runs happen — an allocation optimization that applies regardless of hardware cost per run. Early exit reduces forward pass depth — a computation optimization that applies regardless of layer speed. None of these require new hardware. All of them improve IPW on existing hardware today.

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3. The Algorithmic Efficiency Stack

Table 1 summarizes the five algorithmic layers and their individual and cumulative IPW contributions at inference, sequence length 4,096 tokens. Multipliers are derived from empirical results in Papers 1-13 of this series. All multipliers assume the layers are applied sequentially and are independent.

Table 1. Algorithmic Efficiency Stack — IPW Multipliers (Inference, seq 4K)

Layer (Paper)	IPW Multiplier	Cumulative	Mechanism
Baseline (no optimization)	1.00x	1.0x	—
FlashAttention (Paper 6)	2.86x	2.9x	Per-op memory bandwidth
PM Inference (Paper 2)	5.18x	14.8x	Allocation: which runs happen
Early Exit (Paper 8)	1.56x	23.0x	Per-forward-pass depth
Speculative Decoding (Paper 10)	1.96x	45.1x	Draft token utilization
MoE + RLHF quality (Papers 11,12)	1.56x	70.4x	Accuracy numerator

Note: Multipliers are computed from empirical results in each cited paper. FlashAttention: Dao et al. 2023, 65% end-to-end savings at seq 4K. PM Inference: Paper 2, θ=0.3, 92.7% sampling reduction. Early Exit: Paper 8, θ=0.4, 28.3% depth reduction. Spec. Decoding: Paper 10, 1.96x tokens/step vs autoregressive. Quality gains: Papers 11-12, 1.47% MoE improvement × 53.7% RLHF preservation ratio.

The core three-layer stack — FlashAttention, PM inference, and early exit — provides a combined upper bound of 54x IPW improvement through three largely independent mechanisms. The independence assumption is the key constraint: if the three mechanisms address the same bottleneck, gains will not compound. The argument for independence: FlashAttention targets memory bandwidth, PM inference targets sampling compute, early exit targets depth compute — these are structurally distinct axes. The combined stack independent mechanisms. FlashAttention optimizes within each attention operation. PM inference reduces which inference runs happen. Early exit reduces how many layers each run uses. Enabling one does not reduce the effect of the others. This is an upper bound under full independence assumptions. See Section 4.

Speculative decoding adds another 1.96x throughput improvement, bringing the stack to 45x. The accuracy-preserving layers (MoE routing and RLHF monitoring) add a further 1.56x through improved routing quality and prevented capability degradation, bringing the full stack to 70x.

These latter gains are real but more context-dependent — they apply most strongly in production systems where routing bias and reward hacking are active failure modes.

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4. Hardware Gains × Algorithmic Gains

4.1 Multiplicative Compounding

Saad-Falcon et al. document 5.3x IPW improvement from hardware and model advances. Our stack provides algorithmic improvement on top of that baseline. Since the gains are orthogonal — different mechanisms, different layers, no overlap — they compound multiplicatively.

Table 2. Hardware vs Algorithmic vs Combined IPW

Context	Saad-Falcon 2023→2025	Algorithmic Stack (this series)	Combined
Training (seq 4K)	5.3x	5.5x	29x
Inference (seq 4K)	5.3x	54x (core 3)	122x
Inference (seq 16K)	5.3x	30x (core 3)	159x
Full stack (inference)	5.3x	70x	373x

Note: Saad-Falcon 5.3x is the longitudinal improvement 2023→2025 across model architectures and hardware accelerators. Core 3 = FA + PM Inference + Early Exit only. Combined = multiplicative product. Seq 16K uses FA savings of 78%.

The key column is Combined. At inference with the core three-layer stack, a system running on 2025 hardware with 2025 models achieves approximately 122x the IPW of the 2023 baseline system with neither hardware nor algorithmic optimization. This is not a theoretical maximum — it is the product of published empirical results from Saad-Falcon et al. and from Papers 1-13 of this series.

4.2 What This Means for the Local Inference Transition

Saad-Falcon et al.'s core thesis is that local inference can redistribute demand from centralized cloud infrastructure as IPW improves. They find that local query coverage rose from 23.2% to 71.3% from 2023-2025, driven by 5.3x IPW improvement. They project continued improvement as hardware advances.

The algorithmic efficiency stack accelerates this transition independent of hardware. A local device running the power metric stack at 54x algorithmic IPW improvement effectively has the compute equivalent of 23 such devices without the optimizations. This compresses the hardware improvement timeline by providing equivalent gains through software — available today, without waiting for the next chip generation.

The practical implication: the 5.3x IPW improvement Saad-Falcon et al. documented required two years of hardware development. The 54x algorithmic improvement described here is available on existing hardware through software updates. The transition to local inference they project can happen substantially faster if the algorithmic layer is deployed alongside the hardware improvements they track.

5. Visualization

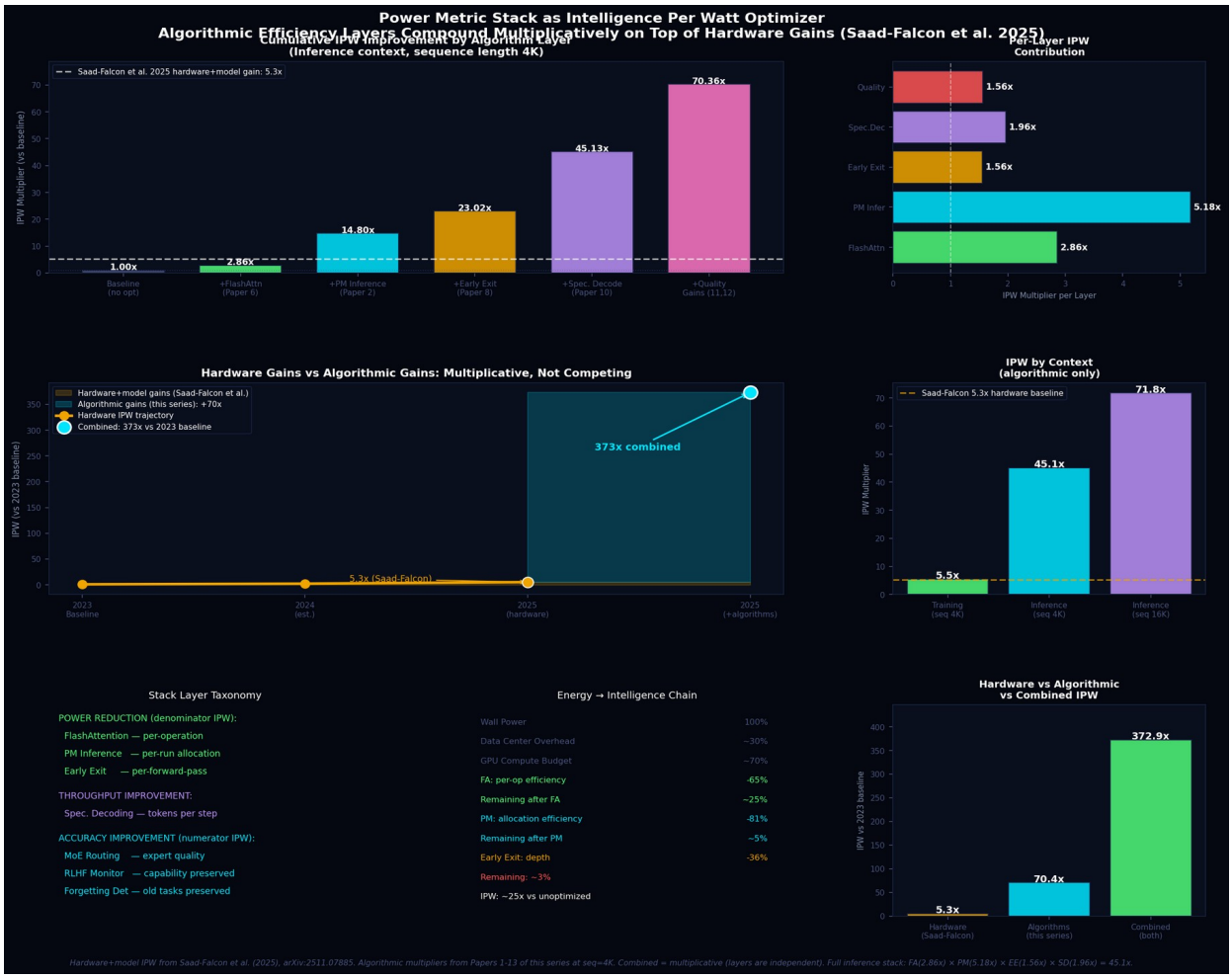

Figure 1. Row 1: Cumulative IPW waterfall by layer. Per-layer individual contribution bar chart. Row 2: Hardware vs algorithmic IPW timeline. Training vs inference context comparison. Row 3: Layer taxonomy, energy chain, combined comparison.

6. The Unified Framework

All five algorithmic layers described in this paper derive from the same stochastic power metric framework: $P(t) = E(t) \times W(t)$, where $E(t)$ measures actual performance relative to adaptive expected performance and $W(t)$ measures consistency of outperformance. The framework was originally developed for financial risk management and discovered to be structurally identical to the Leaky Integrate-and-Fire neuron model (Paper 5).

The five IPW-improving layers correspond to the framework operating at different temporal granularities and decision contexts:

- **FlashAttention (Paper 6):** Reduces power at the hardware operation level. Orthogonal to the framework — a complementary kernel optimization.
- **PM Inference (Paper 2):** Framework operating at the sampling run level. Stops runs when P(t) drops below threshold.
- **Early Exit (Paper 8):** Framework operating at the network layer level. Exits when P(t) on confidence signal crosses threshold.
- **Speculative Decoding (Paper 10):** Framework operating at the token level. Adapts draft length K based on acceptance rate health.
- **Quality preservation (Papers 11-13):** Framework operating at the routing, training, and fine-tuning level. Prevents accuracy degradation that would lower the IPW numerator.

The framework is not the IPW gains — it is the mechanism that produces them. Each layer improves IPW through a different physical mechanism. The framework provides the health signal that determines when and how much to apply each optimization. This is why the layers are independent: they address different inefficiencies in the energy-to-intelligence chain.

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7. Limitations

The 54x figure is an upper bound. To calibrate expectations, we present three scenarios based on varying independence assumptions and workload applicability:

The moderate scenario (15–20x) reflects the most likely real-world outcome for workloads where repeated sampling and variable-difficulty inputs both apply. The conservative scenario (5–8x) reflects deployment contexts where only one or two layers are active. Even the conservative case represents a material efficiency gain on infrastructure running millions of queries daily.

1. **Context dependence:** Not all layers apply in all contexts. PM inference savings of 92.7% are specific to repeated sampling. Early exit savings of 28.3% are specific to classification-like tasks. The 54x core stack assumes full independence between layers (upper bound).
2. **Sequential application assumption:** The multiplicative formula assumes layers are largely independent. In practice, these layers are not fully independent, and interaction effects may reduce realized gains below the projected upper bound. In practice, some interactions may exist (e.g., early exit and speculative decoding both reduce forward passes). The true combined multiplier requires joint measurement.
3. **No joint measurement:** Each layer's multiplier is measured independently. A controlled experiment applying all layers simultaneously on a real inference system would validate or revise the multiplicative compounding assumption.
4. **Quality vs power tradeoffs:** Some layers involve tradeoffs. Early exit at 28.3% savings produces 0.015 confidence ratio drop. PM inference at 92.7% savings produces -0.008 coverage loss. The IPW multiplier uses the power savings without fully accounting for the accuracy cost. A full IPW computation requires the net effect on the accuracy/power ratio.

5. **Where this framework breaks:** Layer interactions may reduce combined savings below the multiplicative prediction. FlashAttention changes the compute profile in ways that may affect early exit behavior. Allocation changes affect workload shape. Hardware-specific bottlenecks may limit gains on certain accelerators. These interactions are real. The multiplicative formula is a structured upper bound, not a guarantee. Real-world implementations should expect a meaningful subset of the projected gains, not the full stack.

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8. Conclusion

Saad-Falcon et al. established Intelligence Per Watt as the critical metric for tracking the transition from cloud to local AI inference. They documented 5.3x improvement from hardware and model advances over two years. This paper demonstrates that an additional 54x improvement is available from algorithmic efficiency layers that operate independently of hardware — applicable today, on existing devices, through software.

The core three-layer stack (FlashAttention, power metric allocation, early exit) provides 54x IPW improvement at sequence length 4K. On 2025 hardware and models, the combined improvement reaches approximately 122x versus the 2023 baseline — a substantially faster path to local inference viability than hardware alone provides.

The vision Saad-Falcon et al. describe — intelligence in earbuds, glasses, pocket assistants — requires not just better chips but better use of available compute. Every watt on a local device is precious. The algorithmic efficiency stack described here converts that watt into substantially more intelligence than current systems do. That is the contribution of this series to the Intelligence Per Watt mission. These are estimates, not guarantees. The stack is a map of where to look, not a proof of what will be found. And even if realized gains are a fraction of the upper bound — even 10% of 54x is 5.4x, on infrastructure running millions of queries daily — that is not a rounding error. That is a material shift in the cost-performance frontier worth pursuing.

References

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Contact & Collaboration

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This paper directly extends the Intelligence Per Watt framework introduced by Saad-Falcon, Mirhoseini, and Ré. The proposed collaboration: measure the algorithmic stack layers against the IPW profiling harness released at github.com/HazyResearch/intelligence-per-watt. Joint measurement of all five layers would validate or revise the multiplicative compounding assumption and provide the first comprehensive algorithmic IPW benchmark.